



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.5

Powers and Exponents

Lesson 6-1 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can write and evaluate numbers in exponent form using a base and a power.

Language Objective: I can explain my work using the words exponent, base, power, and evaluate.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Exponent

Base

Power

Evaluate



Tap any word to see what it means and a picture.

1

The small raised number that shows how many times the base is multiplied is the

2

The number being multiplied repeatedly in a power, like the 3 in 3^4 , is the

3

A number written with a base and an exponent, like 3^4 , is a

4

To find the value of an expression, like $2^3 = 8$, is to

it.

Reflect — Exit Ticket

What is the value of 3^4 ?

- A. 81
- B. 12
- C. 34
- D. 64

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Exponent
2. **Notes 2:** Base
3. **Notes 3:** Power
4. **Notes 4:** Evaluate
5. **Watch for this mistake:** *A common mistake in Powers and Exponents is skipping the key idea: "A power like 2^3 is a short way to write repeated multiplication: $2 \times 2 \times 2$." — always check your work against this rule before you submit.*
6. **Try It 1:** A. 7^3 — *The base 7 is used as a factor 3 times, so the power is 7^3 . $7^3 = 7 \times 7 \times 7 = 343$.*
7. **Try It 2:** A. 64 — $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$. *The exponent 6 counts how many times 2 is a factor.*
8. **Exit Ticket:** A. 81 — $3^4 = 3 \times 3 \times 3 \times 3 = 81$.